

Lab 3 - projectile motion

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Learning Outcomes

1. Learn how to use kinematics equations to make predictions based on a physical model.
2. Develop skills in creating and performing scientifically sound experiments that make experimental determinations and predictions of unknown quantities.
3. Use statistical methods, including mean, standard deviation of the mean, and standard error, to analyze the results of a set of measurements.

Materials and Apparatus

1. spring gun
2. "projectile"
3. carbon paper
4. plain paper
5. measuring stick
6. class laptop, data interface, and photogate sensors
7. googlesheet for data table and analysis

Safety Awareness

Be aware of your surroundings when working with the spring gun and the projectile. The projectile may move extremely fast and cause injury if it strikes someone. Make sure everyone in the vicinity of the launcher is out of the projectile path and aware of each pending launch.

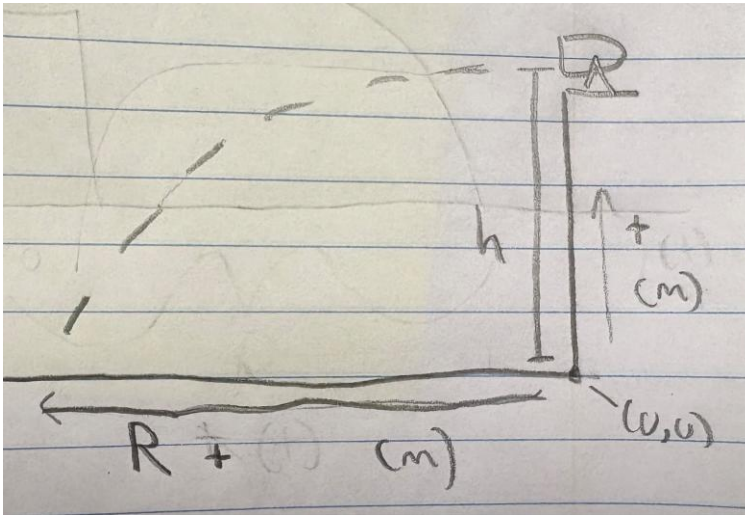
Activity – Determining launch speed.

The purpose of this activity is to determine the launch speed of a projectile. This launch speed may be used for a follow-up activity.

Set-up

1. Mount the spring gun at a vertical distance h above the target that the ball will strike.
2. Set up the computer, sensors, and data interface so that you are able to record the ball's launch speed.

3. Sketch your group's setup and label the origin. Use x-y axes to indicate (+/-) directions.



$$h = 120.0\text{cm} (1.200\text{m})$$

Acquire the Data

You will be collecting your data in two different ways.

Note: Read **Method 1** and **Method 2** before beginning your data runs. You will need to record data for **both** methods for **each** shot you make.

Method 1:

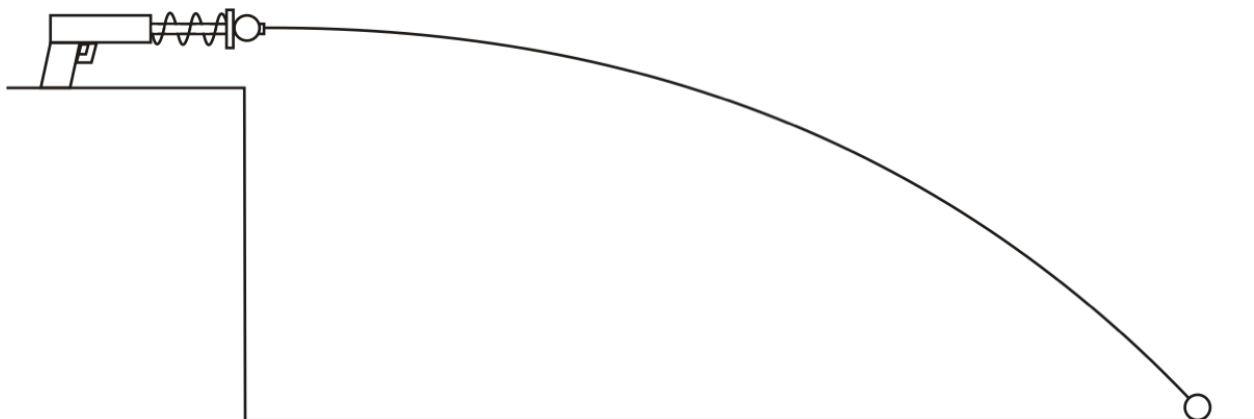
Use your data interface and photogate sensors to **measure the launch speed** for each shot. As directed by the instructor, set up the interface so that you can use the sensor to measure the launch speed for each trial. If the software is able to allow you to measure the launch speed to a variable number of decimal places, then set it to display its data to at least four decimal places (if possible). **Record this data in a data table. You will repeat this for 10 trials ($n = 10$).**

Data Table:

Range (m)	speed capstone (m/s)
1.580	2.7544
1.510	2.6468
1.559	2.6944
1.566	2.7076
1.496	2.5725
1.561	2.7760
1.634	2.8300
1.635	2.7838
1.611	2.8252
1.582	2.6944

Method 2:

To determine the launch speed of a cannonball, you will fire a cannonball horizontally from a height h and observe the ball hit the ground R meters away as measured from the base of the cannon.



PRELAB 1 (show work in notes then insert a screenshot for the group submission):
Find an expression for the cannonball's launch speed v_0 in terms of h , g , and R only.

Handwritten notes on lined paper showing the derivation of the launch speed v_0 in terms of height h , gravity g , and range R .

Vertical motion (y)
 $v_{y0} = 0$
 Vertical acceleration = g

$h = \frac{1}{2} g t^2 \rightarrow$ solve for time of flight t :
 $t = \sqrt{\frac{2h}{g}}$

Horizontal motion
 - constant velocity
 $R = v_0 t$
 - substitute for t

$R = v_0 \sqrt{\frac{2h}{g}} \rightarrow$ solve
 $v_0 = R \sqrt{\frac{g}{2h}}$

Measure and record all the data that you need in order to determine the launch speed. The minimum data you need to collect is 1 value of h and 10 values of R .

Notes:

- To measure R , you need to fire the ball several times and measure the horizontal distance from the launching point to where the ball hits the surface. To do this, In a trial run, fire the ball horizontally, observing the approximate location where the ball hits the surface. Then, place a sheet of carbon paper on top of a sheet of plain paper and center the two sheets on the point where the ball landed during its trial run. You will use these sheets to record and measure the location where the ball hits the surface. (Only the bottom sheet needs to be taped to the surface. **DO NOT** tape the carbon paper to anything.)
- Reset the gun, being careful that it is positioned in a fixed location. Fire the ball horizontally **a minimum of ten times**. Measure and record the horizontal distance R from the ball's launch point to where it hit the surface for each trial. **Record this data in a data table.**

Analysis

PRELAB 2: Read and review [Basic Data Analysis-2A](#) (also in Module 0).

Method 1

- a. Report the initial launch speed, v_0 , for each individual trial with 4 decimal places as measured by the photogate.

speed (m/s)

1. 2.7544
2. 2.6468
3. 2.6944
4. 2.7076
5. 2.5725
6. 2.7760
7. 2.8300
8. 2.7838
9. 2.8252
10. 2.6944

- b. Use these values to calculate the ball's mean launch speed $\langle v_0 \rangle$.

(Add all values and divide by 10)

Average Speed: 2.7285 m/s

- c. Calculate the standard deviation σ for the ball's mean launch speed. The mean plus its standard deviation gives you an estimate of the range of launch speeds for a "typical" shot. Report the estimate of the typical launch speed of a ball for any trial as $v_0 = \langle v_0 \rangle \pm 2\sigma$.

Standard Deviation: 0.0812 m/s

Method 2

- a. Using the data you collected for R and h and your equation on page 2 relating these to v_0 , calculate v_0 for each individual launch.

$$h = 1.20 \text{ m}, \quad v_0 = R\sqrt{2h/g}$$

Range R (m)	Calculated V_0
1.58	3.192
1.51	3.051
1.559	3.150
1.566	3.164
1.496	3.023
1.561	3.154
1.634	3.302
1.635	3.304
1.611	3.255
1.582	3.196

- b. Use these values to calculate the ball's mean launch speed $\langle v_0 \rangle$.

Mean Launch Speed: 3.1793 m/s

- c. Calculate the standard deviation σ for the ball's mean launch speed. The mean plus its standard deviation gives you an estimate of the range of launch speeds for a "typical" shot. Report the estimate of the typical launch speed of a ball for any trial as $v_0 = \langle v_0 \rangle \pm 2\sigma$.

Standard Deviation: 0.0938 m/s

Compare the two values of the typical launch speed of a fired ball measured with the two different methods by calculating the %-difference of the averages. How close do the values match?

$$\% \text{ difference} = (|A - B|) / (B) \times 100\%$$

$$|3.1792 \text{ m/s} - 2.7285 \text{ m/s}| / (2.7285 \text{ m/s}) \times 100\% = 16.52\%$$

The values have a percent difference of 16.52%.

Why did I ask you to calculate the %-difference and not the %-error?

Percent error requires a known true value, but the 2 methods do not give us the true launch speed (Both are experimental measures).

The % difference compares the two measured values without assuming either is correct

Is the typical launch speed calculated by Method 2 within the range of the typical launch speed calculated by Method 1? What about the typical launch speed calculated by Method 1? Is it within the range of the typical launch speed calculated for Method 2? Explain these results.

Method 1: Mean = 2.7285 m/s +/- 0.0813 m/s

Method 2: Mean = 3.1792 m/s +/- 0.0938 m/s

The launch speeds appear not to overlap, which could suggest that the capstone application runs slightly slower. Another reason could be that our cannon was not perfectly level for every launch, which could lead to some variation.

More variation for the error could be air resistance, alignment error, or measurement uncertainty in R or h.

Part 2 – Hitting a Target

If you were to fire a ball at a certain angle θ , how far away must you place a target such that the ball has a reasonable chance of hitting it?

$$R = ((V_0)^2(\sin(2\theta))) / g$$

Set-up

- 1) Set the launch angle for the cannon to a random angle $0^\circ < \theta < 90^\circ$ from horizontal.
45 degrees

Stop and Think

Do you think the uncertainty in the ball's launch speed will influence the uncertainty in its

range? Why or why not?

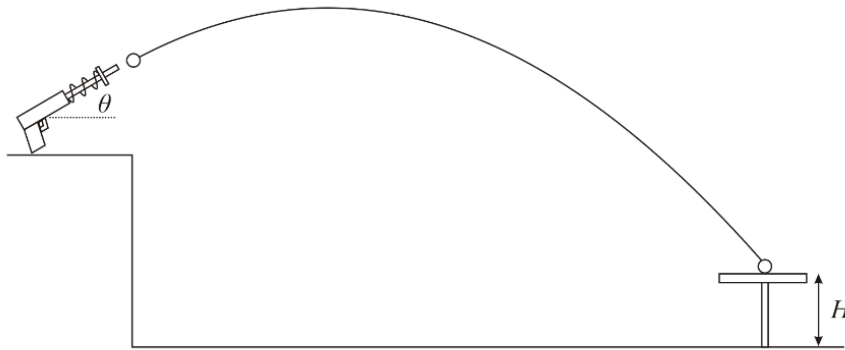
Uncertainty in a ball's launch speed will directly influence uncertainty in range. Since range is calculated with the ball's velocity any changes in velocity directly change the range's outcome.

Which launch speed from the previous part will you use? Why?

We will use the slow speed in order to keep the data consistent.

If a cannonball is fired from an initial height of h meters above the ground and is launched with a speed of v_0 at an angle of θ above the horizontal, where will the ball land? Express your answer for R' in terms of g , h , v_0 , and θ .

$$R = \frac{(v_0)^2 \sin(2\theta)}{g}$$



Acquire the Data

Use your answer to the previous question and the data for launch speed acquired in the previous lab to predict how far away you must place the target such that the ball has a reasonable chance of hitting it. Your prediction should indicate the range of horizontal distance where the ball will fall most of the time. A good way starting point is to do a range of 2 standard deviations away from the center on either end (so the ball may fall anywhere between two standard deviations less than or two standard deviations more than the mean score). In a normal distribution this should usually cover about 95% of the values.

Handwritten calculation for the range R of a projectile:

$$R = \frac{v_0^2 \sin(2\theta)}{g}$$

Given: $v_0 = 3.17 \text{ m/s}$, $\theta = 45^\circ$, $g = 10 \text{ m/s}^2$

$$R = \frac{3.17^2 \sin(2 \cdot 45^\circ)}{10}$$
$$= \frac{10.0489 \sin(90^\circ)}{10} = 0.89 \text{ m}$$
$$\approx 1 \text{ m}$$

Final result: $R \approx 1 \text{ m}$

- On a sheet of paper, draw a circle with a diameter equal to your range on the previous question. Place this on the spot on the ground where you predict the ball will land.
- Do a couple of practice shots to see where the ball lands. Don't worry if it doesn't hit your targets. That's part of the experiment!
- Once you have a good idea where the ball will likely land, place a sheet of plain paper (taped to the ground) and carbon paper at the spot where the ball will have a reasonable chance of striking it a number of times.
- Test your predictions by firing the ball at least 10 times and measuring the distance from the launch point to where the ball lands for each shot.

From these data, calculate the ball's mean range and the standard deviation of its mean range.

45degree ->	Range (m)	speed capstone (m/s)		Range (m)	Calculated V ₀
	1.734	3.0669		1.734	3.503938784
	1.788	3.0555		1.788	3.613057985
	1.616	2.9289		1.616	3.265493123
	1.642	2.9271		1.642	3.318031997
	1.681	2.8349		1.681	3.396840309
	1.637	2.9782		1.637	3.307928367
	1.616	2.9393		1.616	3.265493123
	1.727	3.1034		1.727	3.489793702
	1.792	3.0612		1.792	3.621140888
	1.655	3.0274		1.655	3.344301434
	Avg Range	Avg of Speed			Avg of V ₀
	1.6888	2.99228			3.412601971
	StDev of Range	Stddev of speed			Stdev of V ₀
	0.06722235078	0.08428456825			0.1358379481
	Sigma of Range	Sigma of Speed			Sigma of V ₀
	0.02125757381	0.02665312073			0.04295573087

POSTLAB: Analysis Questions

1. Compare your prediction to the measured results by calculating a %-difference between the predicted value of the range and the measured value of the ball's range.

NOT NEEDED

2. How clustered are the spots where the ball hit? That is, are most of the spots grouped tightly around a central point, or are they more spread out? The level of clustering of the spots where the ball hit may be used as a measure for the experiment's precision. *The more tightly clustered the data, the higher the precision of the experiment.*

A little over half the shots landed close together with a few outliers falling short.

3. Did the ball land at or around a consistent location, even if it wasn't on the predicted target? Did it consistently land short of the predicted target, farther than the target, or centered right around the target? The location of the cluster may be used as a measure of the experiment's accuracy. If data shows that the ball consistently fell short or consistently landed too far away from the predicted target, one may suspect the presence of systematic uncertainty in the experiment. *If the data is clustered around the predicted target (the target's center), then we would consider the experiment to be accurate.*

The ball landed precisely on the plain sheet of paper, but it was not accurate because the cluster of balls that landed on the sheet of paper did not hit our predicted target.

Judging from the impact marks left by the ball on the paper, was your experiment precise? Was it accurate? Do you think the experiment was more precise than accurate, or was it more accurate than precise? Discuss your reasoning.

It was precise, and it was more precise than accurate. The marks on the paper were located in a general area but it was not in the center of the paper meaning it was not accurate but it was precise.

Does the location of the ball's marks indicate the presence of random error, systematic error, or both when you compare it to the predicted target? Which type of error, random or systematic, do you think had a greater influence on the accuracy of this experiment? Which type of error, random or systematic, do you think had a greater influence on the precision of the experiment? Why?

Systematic errors had a greater impact on the experiment's accuracy because they shifted the entire cluster of marks away from the true value. In addition, Capstone played a role in the accuracy of the experiment because, on multiple trials, it failed to record the proper data. Random error had a greater influence on the precision of the experiment because it determines how spread out the marks were from the mean range.

Part 1		
Range (m)	speed capstone (m/s)	Calculated V ₀
1.58	2.7544	3.192
1.51	2.6468	3.051
1.559	2.6944	3.150
1.566	2.7076	3.164
1.496	2.5725	3.023
1.561	2.776	3.154
1.634	2.83	3.302
1.635	2.7838	3.304
1.611	2.8252	3.255
1.582	2.6944	3.196
Avg of Range	Avg of Speed	Avg of V ₀
1.573	2.72851	3.179256654
StDev of Range	Stddev of speed	Stdev of V ₀
0.046457149	0.081248322	0.093849712
Sigma of Range	Sigma of Speed	Sigma of V ₀
0.01469104	0.025692975	0.029677885

Part 2	45degree ->	Range (m)	speed capstone (m/s)	Range (m)	Calculated V ₀
		1.734	3.0669	1.734	3.504
		1.788	3.0555	1.788	3.613
		1.616	2.9289	1.616	3.265
		1.642	2.9271	1.642	3.318
		1.681	2.8349	1.681	3.397
		1.637	2.9782	1.637	3.308
		1.616	2.9393	1.616	3.265
		1.727	3.1034	1.727	3.490
		1.792	3.0612	1.792	3.621
		1.655	3.0274	1.655	3.344
	Avg Range	Avg of Speed		Avg of V ₀	
	1.6888	2.99228		3.413	
	StDev of Range	Stddev of speed		Stdev of V ₀	
	0.067222351	0.084284568		0.135837948	
	Sigma of Range	Sigma of Speed		Sigma of V ₀	
	0.021257574	0.026653121		0.042955731	